PREDICTING STEAM GAME RECEPTION

MINI-PROJECT REPORT BY MIGUEL CORTES

PROJECT OVERVIEW

This project investigates whether a Steam game will be well received by players, defined by a Wilson Score of 0.75 or higher. The Wilson score is a statistically robust measure to review quality that accounts for the total number of reviews, giving a more reliable estimate for games with fewer reviews than a simpler statistical method would.

The central research question is. Can observable game attributes, such as prices, genre, tags, platform support, and playtime, predict whether a game will be well received by the Steam community? Answering this question has practical value for developers when deciding how to position a new title.

DATA WRANGLING AND FEATURE ENGINEERING

The raw dataset contained 122,610 Steam Games. After cleaning and filtering, 40,194 remained. The following steps were applied.

1. Drop irrelevant columns by removing developers, publishers, screenshots, movies, user score, score rank. These are not useful and do not provide us with meaningful data since these are strings that I cant do any analysis on or these are empty without any data.

2. Filter reviews kept only games with more than 20 reviews since there are a lot of asset flips and games with no buyers. Also filtered out adult games by tags and genres. This was to keep this Project school friendly.

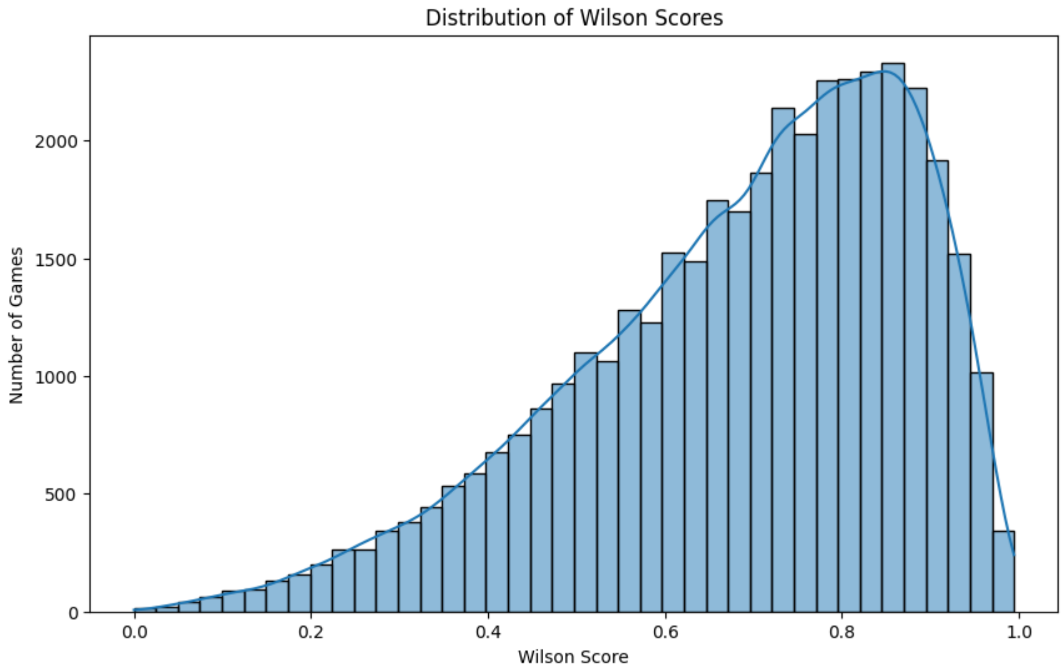
3. Wilson scores from positive/negative review counts. Made binary at 0.75. Provides a measure of reception that accounts for lower review games.

4. Created num_languages, estimated_owners_mid, one-hot encoded genres, and categories. Extracted common tags.

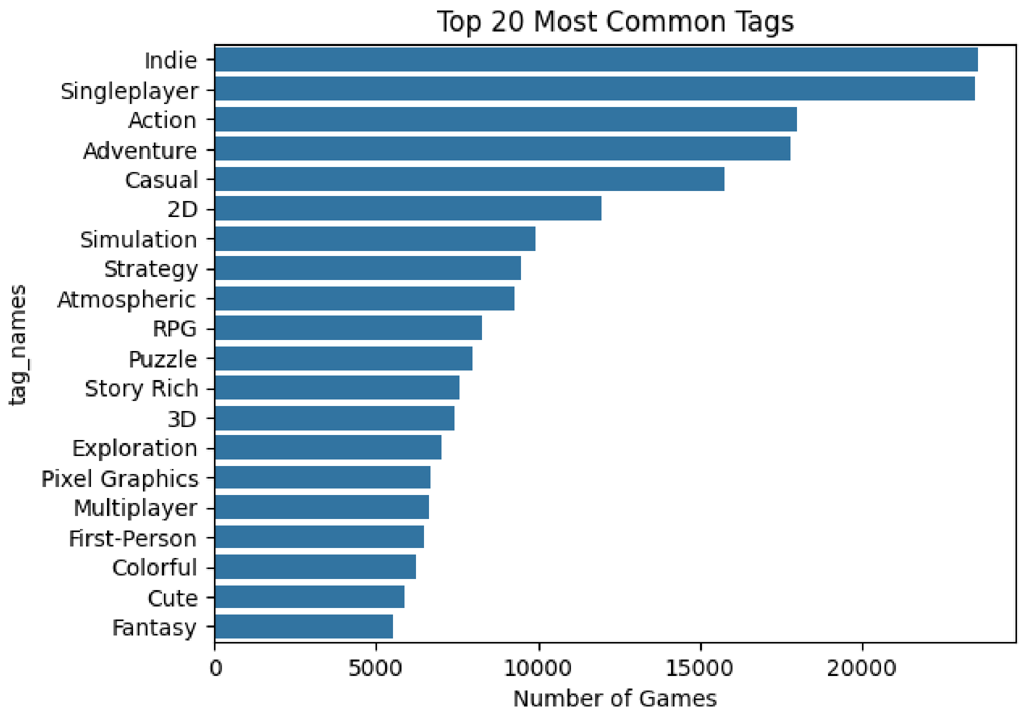
EXPLORATORY DATA ANALYSIS

	count	mean	std	min	25%	50%	75%	max
required_age	40194.00	0.31	2.23	0.00	0.00	0.00	0.00	18.00
price	40194.00	5.30	9.85	0.00	0.99	2.99	6.99	299.90
dlc_count	40194.00	1.12	24.81	0.00	0.00	0.00	1.00	3703.00
windows	40194.00	1.00	0.01	0.00	1.00	1.00	1.00	1.00
mac	40194.00	0.26	0.44	0.00	0.00	0.00	1.00	1.00
linux	40194.00	0.17	0.38	0.00	0.00	0.00	0.00	1.00
achievements	40194.00	33.68	231.13	0.00	0.00	13.00	30.00	9821.00
positive	40194.00	2694.50	46819.88	0.00	30.00	77.00	329.00	7642084.00
negative	40194.00	432.18	9017.30	0.00	8.00	21.00	81.00	1173003.00
average_playtime_forever	40194.00	493.00	17482.66	0.00	0.00	9.00	233.00	3429544.00
median_playtime_forever	40194.00	403.50	17444.86	0.00	0.00	9.00	216.00	3429544.00
discount	40194.00	29.41	33.30	0.00	0.00	0.00	60.00	100.00
peak_ccu	40194.00	130.86	6281.56	0.00	0.00	0.00	2.00	1013936.00
total_reviews	40194.00	3126.68	54475.43	21.00	40.00	102.00	423.00	8815087.00
wilson_score	40194.00	0.69	0.19	0.00	0.57	0.72	0.84	0.99
num_languages	40194.00	5.48	9.58	0.00	1.00	2.00	8.00	103.00
estimated_owners_mid	40194.00	207748.05	2057811.74	10000.00	10000.00	10000.00	75000.00	150000000.00
genre_count	40194.00	2.89	1.33	0.00	2.00	3.00	4.00	16.00
tag_count	40194.00	14.61	6.31	1.00	9.00	18.00	20.00	21.00

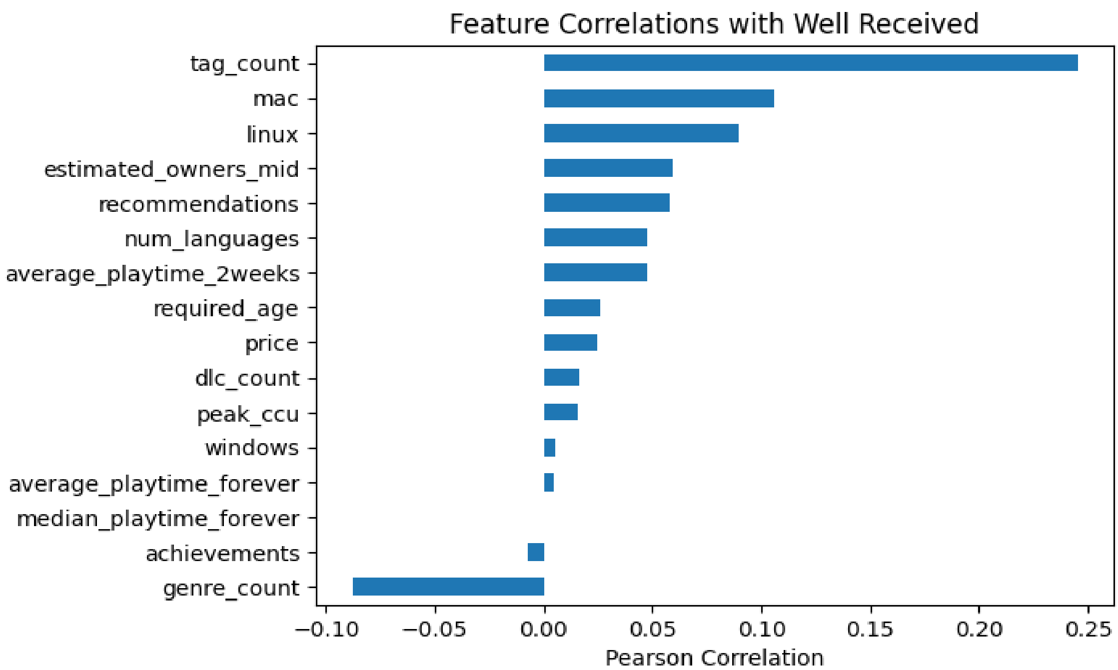
Here we have the descriptive statistics table that shows information about the integer variables in my data frame.



Here we see the distribution of Wilson scores is skewed to the left, where most games are on 0.6 to 0.9 on the Wilson score. I would have thought this would have been a normal distribution, but the reason would be that people are more willing to leave a positive review if they like a game than if they didn’t. Something suggesting this would be that the mean negative review count is 432.18 and the mean positive review is 2694.50, the positive is over 6 times the negative.



Above we have the top twenty most common tags. The two most common would be "indie" and "single player" tags. This makes sense since indie games make up many games, since "indie" means not made by a big studio. Which is easier to make. Single player also makes sense since a game can either be multiplayer or single player, with single player being easier for indie devs to make games for since it takes less knowledge on networking and need to provide servers if they are not using a peer-to-peer protocol.



Here we have correlation of variables with well received. The topmost is tag count with second and third being Linux and mac. Tag count might be so high since they are user rated; users would only vote on tags if it's a game they like. Mac and Linux are both operating systems that only about 20 percent of games support these OS. The fact that it has a high correlation to being well received might be that when a game is doing well enough the devs might want to port it to other OS's.

STATISTICAL QUESTION

Is there an association between a game's Steam tag and whether it is well received? For this, I used a chi-square test to determine whether a distribution of well-received games differed across primary Steam tags. I used the chi-square test since the variables are categorical. Cramer's V was calculated to measure the strength of the association.

The chi-square value is 943.9241 with 19 degrees of freedom and a p-value < 0.0000, indicating a statistically significant association between the Steam tag and whether a game will be well-received. Cramer V was 0.179, indicating a weak to moderate association between the variables.

In conclusion, there is a statistically significant relationship between a game's primary Steam tag and whether it is well received. It is not a strong standalone predictor.

PREDICTIVE MODELLING

A logistic regression classifier was trained to predict whether a game will be well received (Wilson score >= 0.75). The feature set includes counts for tags, genre count, price, one-hot encoding for genres, tags, and categories (tags used had to appear in more than five thousand games), and more. Yielding 507 total features. I tuned the parameter max_iter to be 1000. I did this to make sure I wouldn't get any warnings and to avoid any potential underfitting. I also picked the best C using the validation data split, the best C I got was C=100 with a val AUC = 0.7189. I did this to use the data split in a meaningful way.

DATA SPLIT

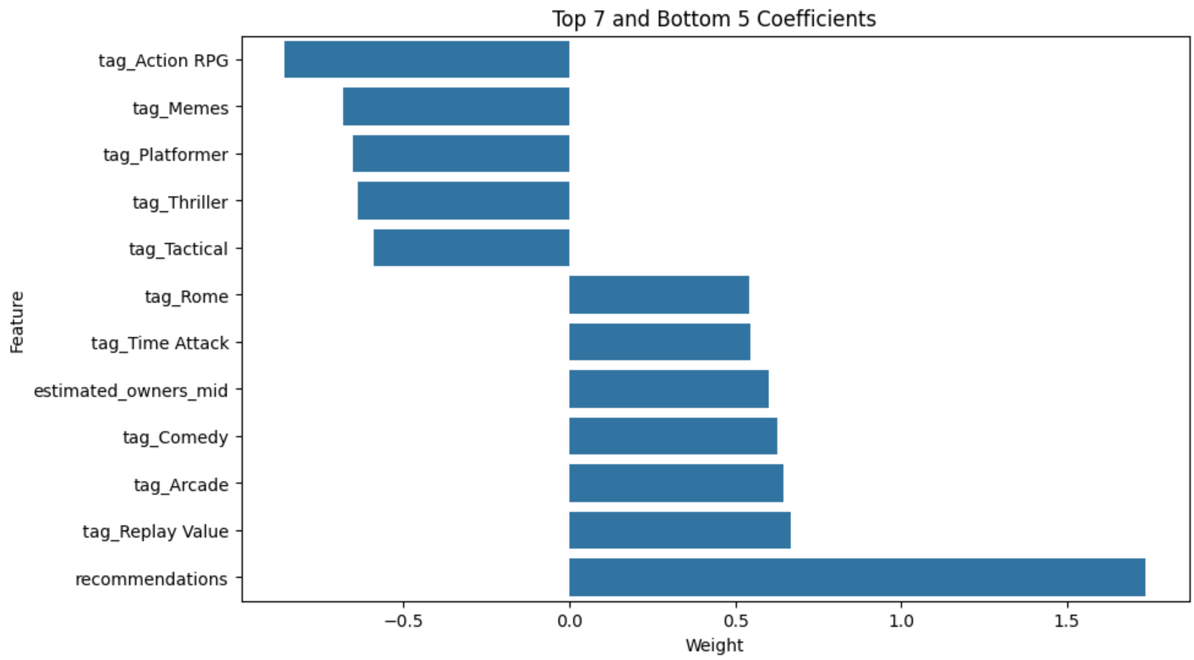
Split	Size	Rows
Training	60%	24,116
Validation	20%	8,039
Test	20%	8,039

LOGISTIC REGRESSION PERFORMANCE

Metric	FALSE	TRUE	OVERALL
Accuracy	-	-	0.67
Precision	0.68	0.65	-
Recall	0.77	0.55	-
F1 score	0.72	0.60	-
ROC-AUC	-	-	0.7253

The model classifies 67% of games correctly. The model is better at identifying games that will not be well received (recall 0.77). This would have to be because it is hard to measure if something will be well received based only on data given by Steam. There are plenty of outside variables that we need to consider.

INTERPRETATION AND INSIGHT



From my interpretation of the research and analysis I did, I am led to believe that although we do have a regression accuracy greater than 50 percent, we could use this as an insight but nothing more than that. Looking at the graph above, we can see the top seven and bottom five coefficients. Recommendation being top makes sense because if someone rates a game positively, then they are going to also recommend the game. While the second is replay value, which means that the game is so fun you can keep playing repeatedly, meaning you enjoy the game more. The bottom features are likely because these are generic tags that likely many games have, to the point where many positive and negative games have these. For a game dev I would say focus on a niche genre and have a game that stands out. One limitation would be that I include features that game devs have no control over, such as Estimated Owners and Recommendation. My next feature I would add would be to be able to see trends over time to see what current players want.

AI USE APPENDIX

I used AI to help guide me on what to do next and help me write some parts of code like the Wilson score (just the formula) and some of the regression training. When I had problems and I was stumped, I also asked what was wrong with my code. Even though I used this to help me write some code, I do want to say I understand and know every line that it wrote. I was not just using it without understanding what it was doing.